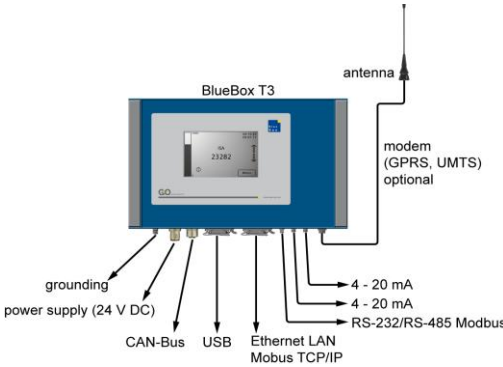
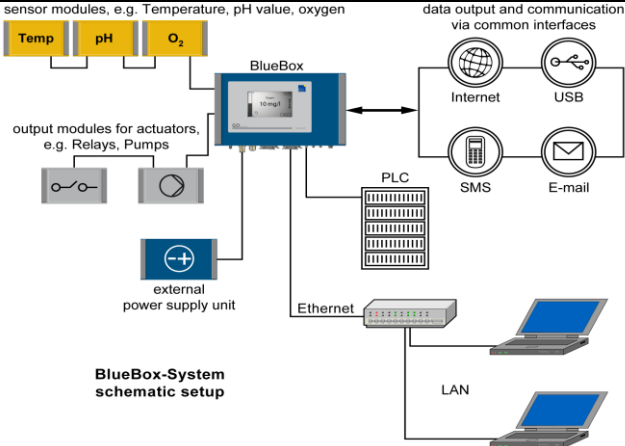
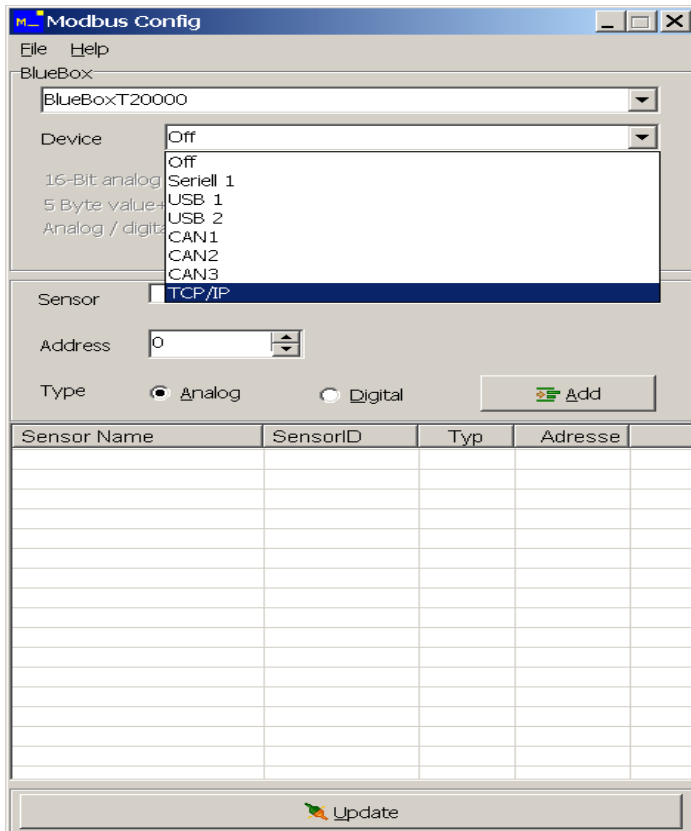
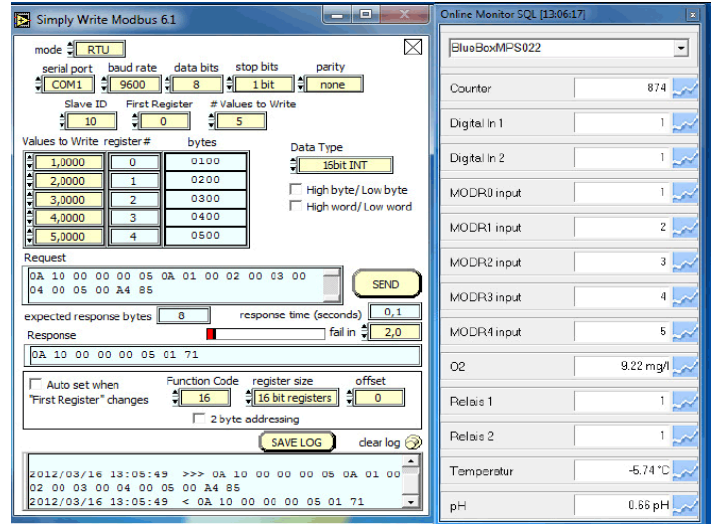
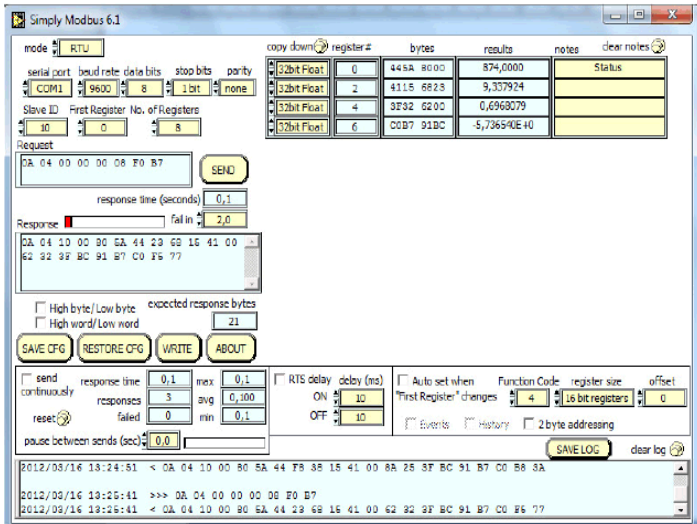
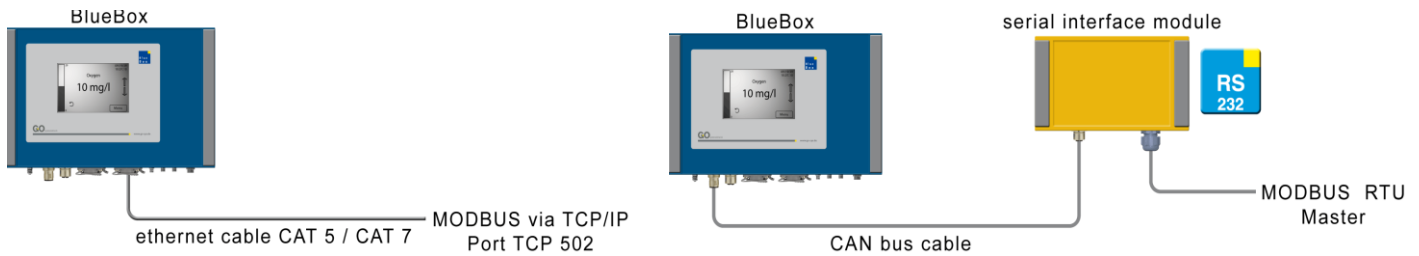


Technical Data:	Bluebox Controller  BlueBox T3 antenna modem (GPRS, UMTS) optional grounding power supply (24 V DC) CAN-Bus USB Ethernet LAN Modbus TCP/IP 4 - 20 mA 4 - 20 mA RS-232/RS-485 Modbus  sensor modules, e.g. Temperature, pH value, oxygen Temp pH O₂ output modules for actuators, e.g. Relays, Pumps external power supply unit BlueBox-System schematic setup data output and communication via common interfaces Internet USB SMS E-mail PLC Ethernet LAN		
Properties	• The BlueBox is the central element of a modular designed measurement and control system. • The BlueBox can be connected with up to 200 sensors and actuators. • Continuous recording • Remote maintenance • In addition virtual sensors (calculated sensor) can be established. • There is a wide range of software.		
Computer	PC 104, 200 MHz to 500 MHz, Operation system Linux®, Random access	Memory	256 MB (optional 512 MB), 1 million result,
Display	Programmable logic controller, Touch-panel, LCD graphic display 240x128 pixel		
Storage media	Industrial Compact Flash Card , 1GB, 2 GB; 512 MB; 256 MB		
Interfaces	2 x mA Outputs; Ethernet 10/100 MBit; CanBus / Modbus; RS 232 / RS 485,		
Logbook	With measuring data of all sensors, calibration history, operation history, error log for more than one year		
Mechanical data	Housing dimensions: 280 mm x 170 mm x 90 mm (W x L x H)	Weight	approx. 2,6 kg
Housing	Die-cast aluminum housing, powder coated	Colour:	RAL 5010
Protection	IP 65, IP66, IP67 (Option)	Power supply	1 x RS-232 or RS-485 (half-duplex), 1 x CAN 1 x Ethernet 10/100MBit 1 x USB optional: 2 x current output 4 mA – 20 mA
Modem	optional: Modem UMTS , GPRS, GSM, (3G) optional: Modem ISDN/analogue	Ambient conditions	Ambient temperature: Storage: -25 °C to +50 °C Operation: -20 °C to +45 °C avoid direct sunlight Relative humidity ≤ 90 % (annual average)
Connection options	The BlueBox has the following connection options: 1. Ground connection 2. Power supply (24 V DC) via a 5-pin M12 panel plug (male) 3. Connection CAN-bus via a 5-pin M12 panel jack, CAN-bus connection to the BlueBox sensor-/actuator modules via 4-wire CAN-Bus connection cable 4. USB connection Please note: The USB port on the BlueBox is provided for data storage and update the firmware. 5. LAN port for LAN cables in standard or cross-link design (see also 3.4). 6. Connection serial interface via a 3-pin M8 panel plug (male), RS-232 or RS-485 7. Connection current output 0/ 4 to 20 mA via a 4-pin M8 panel jack (female), record frequency 1s-3600s 8. Hole for an antenna or cable modem ,		
LAN-connection	• The LAN port allows the connection of the Blue Box to a Local Area Network or directly to a PC. • Thus, a retrieval of data and direct communication with the BlueBox is possible. This requires the installation of the BlueBox PC software on your local computer (see Manual BlueBox PC Software). • The connection to a network is made by a RJ-45-connector at the bottom of the BlueBox (see also 3.2 Con-nection options). For this connection you have to choose one of two • different connecting cables. • For connecting to a network you need a standard RJ-45 cable. • For connecting to a PC you need a RJ-45-crosslink cable. (Only if the PC has no automatic switch of the network interface.)		
Connection of Sensors Actuators: Up to 200 Sensors and Actuators can be additionally be connected with the Controller via CanBus or Wireless connection. (i.e. pH, Conductivity, DO, Temp, Flow, Pressure, etc....), With this the BlueBox can work as a monitoring System or a Control system for controlling a process i.e. opening/closing valves, controlling pumps, or sending information via Internet, email or SMS,i.e. in case of alarm etc.			



Ethernet via TCP/IP

- The MODBUS interface is only accessible from the private network addresses:
- 10.0.0.0/8 (10.0.0.1 - 10.255.255.254)
- 172.16.0.0/12 (172.16.0.1 - 172.31.255.254)
- 192.168.0.0/16 (192.168.0.1 - 192.168.255.254)

